PRo3D

Test Protocol

Version 1.0

Purpose: Manual feature verification against the PRo3D Short User Manual

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Reference Manual:	PRo3D Short User Manual v5.5.0
Scope:	All features described in the Short User Manual

Tester Sign-off

Name	Signature	Date

How to Use This Test Protocol

This document provides step-by-step test cases covering all features described in the PRo3D Short User Manual. Each test case specifies:

- **Prerequisites** – what must be set up before starting the test.
- **Steps** – the exact sequence of actions (buttons to click, keys to press).
- **Expected Result** – what the tester should observe if the feature works correctly.
- **Result** – a checkbox the tester fills in as ✓ Pass or ✗ Fail.
- **Notes** – space to record defects, screenshots, or observations.

Notation used in this document:

- **"Button Name"** – a button in the GUI that the tester should click.
- *Menu* → *Item* – a menu path to follow.
- KEY – a keyboard key to press.
- CTRL+LMB – hold Ctrl and click the Left Mouse Button on the 3D surface.
- **Tab Name tab** – a tab in the right-hand side panel.

A test case **passes** when all expected results are observed. If any expected result is not observed, mark the test as **fail** and add details in the Notes field.

Test Summary

Test ID	Feature	Result	Notes
TC-1.1	Start PRo3D	☐ Pass / ☐ Fail	
TC-1.2	Add Surface (OPC)	☐ Pass / ☐ Fail	
TC-1.3	Surface FlyTo	☐ Pass / ☐ Fail	
TC-1.4	Save Scene	☐ Pass / ☐ Fail	
TC-1.5	Load Scene (Open)	☐ Pass / ☐ Fail	
TC-1.6	Load Scene (Recent)	☐ Pass / ☐ Fail	
TC-2.1	FreeFly Navigation	☐ Pass / ☐ Fail	
TC-2.2	ArcBall Navigation	☐ Pass / ☐ Fail	
TC-2.3	Pick Explore Center	☐ Pass / ☐ Fail	
TC-2.4	Place Coordinate System	☐ Pass / ☐ Fail	
TC-3.1	Draw Point Annotation	☐ Pass / ☐ Fail	
TC-3.2	Draw Line Annotation	☐ Pass / ☐ Fail	
TC-3.3	Draw Polyline Annotation	☐ Pass / ☐ Fail	
TC-3.4	Draw Polygon Annotation	☐ Pass / ☐ Fail	
TC-3.5	Draw Dip and Strike Annotation	☐ Pass / ☐ Fail	
TC-3.6	Annotation Projection Modes	☐ Pass / ☐ Fail	
TC-3.7	Pick Annotation (via UI list)	☐ Pass / ☐ Fail	
TC-3.8	Pick Annotation (via 3D view)	☐ Pass / ☐ Fail	
TC-3.9	Pick Surface (via UI list)	☐ Pass / ☐ Fail	
TC-3.10	Pick Surface (via 3D view)	☐ Pass / ☐ Fail	
TC-4.1	Surface Properties Panel	☐ Pass / ☐ Fail	
TC-4.2	Surface Visibility Toggle	☐ Pass / ☐ Fail	
TC-4.3	Surface Color Correction	☐ Pass / ☐ Fail	
TC-4.4	Surface Translation	☐ Pass / ☐ Fail	
TC-4.5	Surface FillMode (Wireframe)	☐ Pass / ☐ Fail	
TC-5.1	Annotation Properties Panel	☐ Pass / ☐ Fail	
TC-5.2	Annotation Measurements	☐ Pass / ☐ Fail	
TC-5.3	Annotation Text Note	☐ Pass / ☐ Fail	
TC-6.1	Place Scalebar	☐ Pass / ☐ Fail	
TC-6.2	Scalebar Properties	☐ Pass / ☐ Fail	
TC-7.1	Add Bookmark	☐ Pass / ☐ Fail	
TC-7.2	Bookmark FlyTo	☐ Pass / ☐ Fail	
TC-8.1	Sequenced Bookmarks – Add & Animate	☐ Pass / ☐ Fail	
TC-8.2	Sequenced Bookmarks – Record & Render	☐ Pass / ☐ Fail	
TC-9.1	Viewer Config Settings	☐ Pass / ☐ Fail	
TC-9.2	Screenshots	☐ Pass / ☐ Fail	
TC-10.1	Grouping – Create Group	☐ Pass / ☐ Fail	
TC-10.2	Grouping – Move Leafs	☐ Pass / ☐ Fail	
TC-11.1	View Planner – Place Rover	☐ Pass / ☐ Fail	
TC-11.2	View Planner – Instrument View	☐ Pass / ☐ Fail	
TC-12.1	GIS View – Load SPICE Kernel	☐ Pass / ☐ Fail	
TC-12.2	GIS View – Observation Settings	☐ Pass / ☐ Fail	
TC-13.1	Contour Lines	☐ Pass / ☐ Fail	
TC-13.2	Multitexturing	☐ Pass / ☐ Fail	
TC-13.3	Cross Sections	☐ Pass / ☐ Fail	

Test ID	Feature	Result	Notes
TC-14.1	Surface Comparison – Activate Mode	☐ Pass / ☐ Fail	
TC-14.2	Surface Comparison – Select Area	☐ Pass / ☐ Fail	
TC-14.3	Surface Comparison – Surface Measurements	☐ Pass / ☐ Fail	
TC-14.4	Surface Comparison – Length Measurements	☐ Pass / ☐ Fail	
TC-15.1	Minerva – Load Data & Listing	☐ Pass / ☐ Fail	
TC-15.2	Minerva – Query App Filter	☐ Pass / ☐ Fail	
TC-15.3	Minerva – Product Selection	☐ Pass / ☐ Fail	
TC-15.4	Minerva – Pick Linking	☐ Pass / ☐ Fail	
TC-16.1	CLI – Snapshot Rendering	☐ Pass / ☐ Fail	
TC-17.1	RIMFAX – Load Traverse	☐ Pass / ☐ Fail	
TC-17.2	RIMFAX – Load Surfaces	☐ Pass / ☐ Fail	
TC-18.1	Keyboard Shortcuts	☐ Pass / ☐ Fail	
TC-19.1	Undo/Redo for Drawing	☐ Pass / ☐ Fail	
TC-19.2	Group Default Color	☐ Pass / ☐ Fail	

Contents

1 Starting PRo3D

TC-1.1: Start PRo3D

Prerequisites	PRo3D is installed.
Steps	1. Double-click `PRo3D.exe` (or the application shortcut). 2. Wait for the application to finish loading.
Expected Result	• The PRo3D main window opens. • A menu icon (hamburger or similar) is visible in the top-left corner of the window. • No error messages or crash dialogs appear.
Result	☐ Pass ☐ Fail
Notes	

TC-1.2: Add Surface (OPC)

Prerequisites	PRo3D is running (TC-1.1 passed). An OPC surface folder is available on disk.
Steps	1. Click the main menu icon in the top-left corner. 2. Hover over the *Import* tab. 3. Click **"OPC"**. 4. In the "Select Folder" dialog, navigate to the OPC folder and click **"Select Folder"**. 5. Wait for the surface to load.
Expected Result	• The "Select Folder" dialog opens and allows folder selection. • After confirming, the surface is listed in the **Surfaces tab** panel on the right side of the window. • The surface name appears in the list.
Result	☐ Pass ☐ Fail
Notes	

TC-1.3: Surface FlyTo

Prerequisites	A surface has been loaded (TC-1.2 passed).
Steps	1. In the **Surfaces tab** panel (right side), locate the loaded surface in the list. 2. Click the **“FlyTo”** button below the surface name.
Expected Result	• The camera performs a smooth fly-to animation. • The 3D surface becomes visible and centered in the main 3D view.
Result	☐ Pass ☐ Fail
Notes	

TC-1.4: Save Scene

Prerequisites	At least one surface is loaded (TC-1.2 passed).
Steps	1. Click the main menu icon in the top-left corner. 2. Navigate to *Scene* → *Save as…*. 3. In the Save dialog, enter a name for the scene file (e.g., `TestScene`). 4. Click **“Save”**.
Expected Result	• The save dialog opens. • After saving, a `.scn` file appears in the chosen directory. • No error messages are shown.
Result	☐ Pass ☐ Fail
Notes	

TC-1.5: Load Scene (Open)

Prerequisites	A scene file (`.scn`) exists on disk (TC-1.4 passed).
Steps	1. Restart PRo3D (or close the current scene). 2. Click the main menu icon in the top-left. 3. Navigate to *Scene* → *Open*. 4. In the file dialog, navigate to the saved `.scn` file and select it. 5. Click "**Open**" (or double-click the file).
Expected Result	• The scene loads successfully. • Previously added surfaces appear in the **Surfaces tab** list. • The 3D view displays the loaded surface data.
Result	☐ Pass ☐ Fail
Notes	

TC-1.6: Load Scene (Recent)

Prerequisites	A scene has been opened at least once before (TC-1.5 passed).
Steps	1. Click the main menu icon in the top-left. 2. Hover over *Scene* → *Recent*. 3. A list of recently opened scenes appears; click the desired scene name.
Expected Result	• A list of recent scene files is shown on hovering. • Clicking a scene name loads that scene without opening a file dialog. • Surfaces from the selected scene appear in the list and 3D view.
Result	☐ Pass ☐ Fail
Notes	

2 Viewer Actions and Navigation

TC-2.1: FreeFly Navigation

Prerequisites	A surface is loaded and visible in the 3D view (TC-1.3 passed). The navigation mode is set to FreeFly.
Steps	1. Ensure the viewer interaction mode is set to **"PickExploreCenter"** (press `F1` or select from the actions menu). 2. Press `W` to move forward. 3. Press `S` to move backward. 4. Press `A` to strafe left. 5. Press `D` to strafe right. 6. Hold `LMB` (Left Mouse Button) and drag to rotate the camera. 7. Scroll the mouse wheel to zoom in and out. 8. Hold `RMB` (Right Mouse Button) and drag to move forward/backward. 9. Hold the Middle Mouse Button and drag to pan.
Expected Result	• `W`/`S` moves the camera forward and backward through the scene. • `A`/`D` strafes the camera left and right. • Dragging with `LMB` held rotates the camera's orientation. • Mouse wheel zooms smoothly in and out. • Dragging with `RMB` held moves the camera forward/backward. • Dragging with Middle Mouse Button pans the view.
Result	☐ Pass ☐ Fail
Notes	

TC-2.2: ArcBall Navigation

Prerequisites	ArcBall navigation mode is active. A surface is loaded.
Steps	1. Switch to ArcBall mode via the navigation mode selector. 2. Hold `LMB` and drag to rotate the camera around the current explore center. 3. Press `W` or scroll the mouse wheel to move forward. 4. Press `S` or scroll the mouse wheel to move backward.
Expected Result	• The camera orbits around the explore center when `LMB` is held and dragged. • `W`/`S` and the mouse wheel change the distance to the explore center. • The explore center remains stationary during rotation.
Result	☐ Pass ☐ Fail
Notes	

TC-2.3: Pick Explore Center

Prerequisites	ArcBall navigation mode is active. A surface is loaded and visible.
Steps	1. In the actions drop-down, select “**PickExploreCenter**” (or press `F1`). 2. Hold `Ctrl` and click `LMB` on a point on the 3D surface. 3. Observe the viewport. 4. Then hold `LMB` and drag to orbit.
Expected Result	• A pink dot appears at the clicked location on the surface. • Orbiting (dragging with `LMB`) now rotates the camera around the new pink dot position.
Result	☐ Pass ☐ Fail
Notes	

TC-2.4: Place Coordinate System

Prerequisites	A surface is loaded and visible.
Steps	1. In the actions drop-down, select “**PlaceCoordinateSystem**” (or press `F4`). 2. Select a unit of measurement from the unit drop-down (e.g., meters). 3. Hold `Ctrl` and click `LMB` on a point on the 3D surface. 4. Observe the viewport and the Config panel.
Expected Result	• A coordinate system widget appears at the clicked position on the surface. • A blue arrow (up vector) points in the positive Z direction. • A red arrow (north vector) points in the positive Y direction. • The position, up vector, and north vector values are updated in the **Config tab** panel under *Coordinate System*.
Result	☐ Pass ☐ Fail
Notes	

3 Drawing and Managing Annotations

TC-3.1: Draw Point Annotation

Prerequisites	A surface is loaded and visible.
Steps	1. In the actions drop-down, select **"DrawAnnotation"** (or press `F2`). 2. In the annotation mode selector, choose **"Point"**. 3. Hold `Ctrl` and click `LMB` on the surface.
Expected Result	• A single point marker appears on the surface at the clicked location. • The new annotation appears in the **Annotations tab** list on the right. • Clicking the annotation name in the list highlights it in green.
Result	☐ Pass ☐ Fail
Notes	

TC-3.2: Draw Line Annotation

Prerequisites	DrawAnnotation mode is active.
Steps	1. In the annotation mode selector, choose **"Line"**. 2. Hold `Ctrl` and click `LMB` on a first point on the surface. 3. Hold `Ctrl` and click `LMB` on a second point on the surface.
Expected Result	• Two points are placed and connected by a line projected onto the surface. • The line appears in the **Annotations tab** list. • The *Measurements* panel shows distance values (Length, WayLength).
Result	☐ Pass ☐ Fail
Notes	

TC-3.3: Draw Polyline Annotation

Prerequisites	DrawAnnotation mode is active.
Steps	1. In the annotation mode selector, choose "**Polyline**". 2. Hold `Ctrl` and click `LMB` to place the first point. 3. Hold `Ctrl` and click `LMB` to place at least two more points. 4. Press `Enter` to finish the polyline.
Expected Result	• Each click places a new point and a connecting line segment is drawn. • Pressing `Enter` finalizes the polyline. • The polyline appears in the **Annotations tab** list. • Measurements (Length, WayLength, Slope, etc.) are shown in the properties panel.
Result	☐ Pass ☐ Fail
Notes	

TC-3.4: Draw Polygon Annotation

Prerequisites	DrawAnnotation mode is active.
Steps	1. In the annotation mode selector, choose "**Polygon**". 2. Hold `Ctrl` and click `LMB` to place at least three points. 3. Press `Enter` to close and finish the polygon.
Expected Result	• Points are placed and connected by segments. • Pressing `Enter` closes the polygon (last point connects to first point). • The polygon appears in the **Annotations tab** list.
Result	☐ Pass ☐ Fail
Notes	

TC-3.5: Draw Dip and Strike (DnS) Annotation

Prerequisites	DrawAnnotation mode is active. A surface with geological structure is loaded.
Steps	1. In the annotation mode selector, choose **"DnS"**. 2. Hold `Ctrl` and click `LMB` to place at least three points along a rock layer. 3. Press `Enter` to finish and compute the dip and strike.
Expected Result	• A polyline (blue) is drawn along the picked points. • A fitted plane is shown with a strike vector (red) and dip vector (green). • A disc visualizing the dip direction appears on the surface. • Dip and Strike measurements (azimuth, dip angle) appear in the properties panel. • The annotation appears in the **Annotations tab** list.
Result	☐ Pass ☐ Fail
Notes	

TC-3.6: Annotation Projection Modes

Prerequisites	DrawAnnotation mode is active in Line or Polyline mode.
Steps	1. In the projection selector, choose **"Linear"**. 2. Draw a line annotation on the surface. Observe the result. 3. Draw another line with **"ViewPoint"** projection and compare. 4. Draw another line with **"Sky"** projection and compare.
Expected Result	• **Linear**: The line is a straight point-to-point connection without surface projection. • **ViewPoint**: The line follows the surface contour as seen from the current camera position; it hugs depressions and ridges. • **Sky**: The line is projected along the scene's up-vector, following the surface from above. • The three lines visibly differ in their path across the surface.
Result	☐ Pass ☐ Fail
Notes	

TC-3.7: Pick Annotation (via UI List)

Prerequisites	At least one annotation exists (TC-3.1 through TC-3.5 passed).
Steps	1. In the **Annotations tab** panel, click on an annotation's name.
Expected Result	• The annotation's name turns green in the list. • The annotation is highlighted with a green border in the 3D view. • The annotation's properties appear in the Properties panel.
Result	☐ Pass ☐ Fail
Notes	

TC-3.8: Pick Annotation (via 3D View)

Prerequisites	At least one annotation exists in the 3D view.
Steps	1. In the actions drop-down, select **"PickAnnotation"** (or press `F3`). 2. Hold `Ctrl` and click `LMB` on or near an annotation in the main 3D view.
Expected Result	• The annotation under the cursor is selected. • It turns green in the annotations list. • It gets a green border in the 3D view. • Its properties are shown in the Properties panel.
Result	☐ Pass ☐ Fail
Notes	

TC-3.9: Pick Surface (via UI List)

Prerequisites	At least one surface is loaded.
Steps	1. Navigate to the **Surfaces tab** panel. 2. Click on the surface's name in the list.
Expected Result	• The surface's name turns green in the list. • The surface's properties appear in the Properties panel.
Result	☐ Pass ☐ Fail
Notes	

TC-3.10: Pick Surface (via 3D View)

Prerequisites	At least one surface is loaded and visible.
Steps	1. In the actions drop-down, select "**PickSurface**". 2. Hold `Ctrl` and click `LMB` on the surface in the 3D view.
Expected Result	• The surface's name turns green in the **Surfaces tab** list. • The surface's properties appear in the Properties panel.
Result	☐ Pass ☐ Fail
Notes	

4 Surface Properties and Controls

TC-4.1: Surface Properties Panel

Prerequisites	A surface is selected (TC-3.9 passed).
Steps	1. With the surface selected, view the Properties panel on the right. 2. Change the surface name by clicking the name field, typing a new name, and pressing `Enter`. 3. Change the *Priority* value to `0.1` and press `Enter`. 4. Change the *FillMode* to `Wireframe`.
Expected Result	• The new name appears in the **Surfaces tab** list. • The priority change is accepted (no error). • The surface renders as a wireframe mesh in the 3D view.
Result	☐ Pass ☐ Fail
Notes	

TC-4.2: Surface Visibility Toggle

Prerequisites	A surface is loaded and visible in the 3D view.
Steps	1. In the **Surfaces tab** list, click **“Toggle Visible”** below the surface name. 2. Observe the 3D view. 3. Click **“Toggle Visible”** again. 4. Observe the 3D view.
Expected Result	• After the first click, the surface disappears from the 3D view. • After the second click, the surface reappears.
Result	☐ Pass ☐ Fail
Notes	

TC-4.3: Surface Color Correction

Prerequisites	A surface is selected and visible (TC-3.9 passed).
Steps	1. With the surface selected, locate the *Color Correction* panel in the Properties section. 2. Adjust the *Contrast* slider. 3. Adjust the *Brightness* slider. 4. Observe the 3D view after each change.
Expected Result	• Adjusting the contrast slider visibly changes the contrast of the surface texture in the 3D view. • Adjusting the brightness slider visibly lightens or darkens the surface texture.
Result	☐ Pass ☐ Fail
Notes	

TC-4.4: Surface Translation

Prerequisites	A surface is selected and a Coordinate System has been placed (TC-2.4 passed).
Steps	1. With the surface selected, locate the *Translation* panel in the Properties section. 2. Enter a small offset value (e.g., `1.0`) in the *North* axis field and press `Enter`. 3. Observe the 3D view. 4. Reset the value back to `0.0`.
Expected Result	• The surface moves by the specified amount along the north axis. • The 3D view updates immediately. • Resetting to 0 returns the surface to its original position.
Result	☐ Pass ☐ Fail
Notes	

TC-4.5: Surface FillMode (Wireframe)

Prerequisites	A surface is selected.
Steps	1. In the surface Properties panel, find the *FillMode* drop-down. 2. Set it to `Wireframe`. 3. Observe the 3D view. 4. Set it to `Point`. 5. Observe the 3D view. 6. Set it back to `Solid`.
Expected Result	• **Wireframe**: The surface is rendered as a mesh of triangles (edges only). • **Point**: The surface is rendered as a point cloud (vertices only). • **Solid**: The surface renders normally with filled triangles and texture.
Result	☐ Pass ☐ Fail
Notes	

5 Annotation Properties and Measurements

TC-5.1: Annotation Properties Panel

Prerequisites	A line or polyline annotation is selected (TC-3.7 passed).
Steps	1. With an annotation selected, find the Properties panel. 2. Change the *Color* to a different color. 3. Change the *Thickness* to a larger value (e.g., 5). 4. Toggle the *Visible* checkbox off, then on. 5. Change the *TextSize* value.
Expected Result	• The annotation updates its color in the 3D view immediately. • The line thickness visibly increases in the 3D view. • Toggling visibility hides and then shows the annotation in the 3D view. • Text size changes affect any label displayed near the annotation.
Result	☐ Pass ☐ Fail
Notes	

TC-5.2: Annotation Measurements

Prerequisites	A polyline annotation with multiple points is selected.
Steps	1. With a polyline annotation selected, navigate to the *Measurements* panel. 2. Read the following values: *Length*, *WayLength*, *Height*, *HeightDelta*, *Slope*, *Bearing*.
Expected Result	• All measurement fields are populated with numeric values. • *Length* shows the straight-line distance sum. • *WayLength* shows the projected distance along the surface. • *Height* shows the elevation difference between start and end. • *Slope* and *Bearing* are displayed in appropriate units (degrees).
Result	☐ Pass ☐ Fail
Notes	

TC-5.3: Annotation Text Note

Prerequisites	An annotation is selected.
Steps	1. In the annotation Properties panel, locate the *Text* field. 2. Type a short note (e.g., `Test Note`). 3. Press `Enter`. 4. Observe the 3D view.
Expected Result	• The entered text appears as a label next to the annotation in the 3D view.
Result	☐ Pass ☐ Fail
Notes	

6 Scalebars

TC-6.1: Place Scalebar

Prerequisites	A surface is loaded. A coordinate system has been placed (TC-2.4 passed).
Steps	1. In the actions drop-down, select "**PlaceScaleBar**". 2. Select an orientation mode (e.g., `Horizontal_cam`). 3. Choose a position (e.g., `Middle`). 4. Enter a length value (e.g., `10`) and select a unit (e.g., `m`). 5. Hold `Ctrl` and click `LMB` on the surface to place the scalebar.
Expected Result	• A scalebar appears on the surface in the 3D view, aligned according to the selected orientation. • The length label (e.g., `10 m`) is shown beside the scalebar. • The scalebar is listed in the **ScaleBars tab** panel.
Result	☐ Pass ☐ Fail
Notes	

TC-6.2: Scalebar Properties

Prerequisites	A scalebar exists (TC-6.1 passed) and is selected.
Steps	1. Click the scalebar name in the **ScaleBars tab** list to select it. 2. Change the *Thickness* value. 3. Change the *Subdivisions* value (e.g., to `5`). 4. Toggle *TextVisible* to hide the label, then show it again. 5. Click the "**Toggle**" button below the scalebar name to hide/show it. 6. Click the "**remove**" button below the scalebar name to delete it.
Expected Result	• Thickness change updates the scalebar line width visually. • Subdivisions change updates the number of black-white segments on the bar. • Toggling text visibility hides/shows the length label. • Clicking Toggle hides/shows the entire scalebar. • Clicking remove deletes the scalebar from the list and the 3D view.
Result	☐ Pass ☐ Fail
Notes	

7 Bookmarks

TC-7.1: Add Bookmark

Prerequisites	A surface is loaded. Navigate to a desired viewpoint in the 3D view.
Steps	1. Navigate to the **Bookmarks tab** page in the right panel. 2. Click the **"+"** button at the top of the bookmarks page.
Expected Result	• A new bookmark entry appears in the bookmarks list. • The bookmark captures the current camera position.
Result	☐ Pass ☐ Fail
Notes	

TC-7.2: Bookmark FlyTo

Prerequisites	At least one bookmark exists (TC-7.1 passed). Move the camera to a different viewpoint.
Steps	1. In the **Bookmarks tab** list, click the "house" icon beside the bookmark's name.
Expected Result	• The camera animates (flies) back to the saved viewpoint. • The 3D view shows the scene from the position saved in the bookmark.
Result	☐ Pass ☐ Fail
Notes	

8 Sequenced Bookmarks

TC-8.1: Sequenced Bookmarks – Add and Animate

Prerequisites	A surface is loaded. Navigate to at least two different viewpoints to capture.
Steps	1. Navigate to the **Sequenced Bookmarks tab** page. 2. Navigate the camera to a first position, then click the **"+"** button (A in the UI) to add a bookmark. 3. Navigate the camera to a second position, then click **"+"** again. 4. In the bookmark list (B), set a *Duration* for the second bookmark (e.g., 3 seconds). 5. In the Animation section (D), click the **Play** button (▶).
Expected Result	• Two bookmarks appear in the sequenced bookmarks list. • Clicking Play starts a camera animation that travels from the first to the second bookmark. • The animation duration corresponds to the set value. • The Pause and Stop buttons become active during animation.
Result	☐ Pass ☐ Fail
Notes	

TC-8.2: Sequenced Bookmarks – Record and Generate Images

Prerequisites	At least two sequenced bookmarks exist (TC-8.1 passed). An output path is set in the Snapshots section (E).
Steps	1. In the *Snapshots* section (E), ensure the *Output Path* is valid. 2. Set an *Image Resolution* (e.g., `1920x1080`). 3. Click the red **Record** button in the Snapshots section. 4. Click the **Play** button in the Animation section (D) to animate the camera. 5. Wait for the animation to complete. 6. Click the red **Stop** button in the Snapshots section. 7. Click the **Generate Images** button (camera icon).
Expected Result	• The Record button changes to a Stop button when recording starts. • After stopping, a JSON file is saved in the output folder. • After clicking Generate Images, PRo3D renders images in the background. • Images (named `img_000.png` etc.) appear in the output folder.
Result	☐ Pass ☐ Fail
Notes	

9 Viewer Configuration

TC-9.1: Viewer Config Settings

Prerequisites	A surface is loaded.
Steps	1. Navigate to the **Config tab** page in the right panel. 2. Adjust the *Navigation Sensitivity* slider. 3. Change the *Arrow Length* value. 4. Enable *Lod Colors* (if available) and observe. 5. Press `Page Up` to increase navigation sensitivity. 6. Press `Page Down` to decrease navigation sensitivity.
Expected Result	• Adjusting the Navigation Sensitivity slider changes how fast the camera responds to input. • `Page Up`/`Page Down` also adjusts sensitivity (the value updates in the Config panel). • Arrow Length change affects the size of direction arrows visible in the scene. • Lod Colors colors the geometry by level-of-detail (different shades visible on the surface).
Result	☐ Pass ☐ Fail
Notes	

TC-9.2: Screenshots

Prerequisites	A surface is loaded and visible.
Steps	1. Navigate to the **Config tab** page. 2. Locate the *Screenshots* section. 3. Set a width and height (e.g., `1920` and `1080`). 4. Select output format `PNG`. 5. Click the **camera icon** button to capture a screenshot. 6. Click the **folder icon** button to open the output folder.
Expected Result	• Clicking the camera icon creates a screenshot file in the output directory. • Clicking the folder icon opens the operating system's file explorer showing the output folder. • The screenshot file (e.g., `img0.png`) is present in the folder and matches the current 3D view.
Result	☐ Pass ☐ Fail
Notes	

10 Grouping

TC-10.1: Grouping – Create Group

Prerequisites	At least two annotations exist.
Steps	1. Navigate to the **Annotations tab** panel. 2. In the listing, right-click or use the group context menu on the “root” group. 3. Select **“Add Group”** from the context menu. 4. Verify a new empty group appears below root. 5. Right-click the new group and select **“Set Active”**.
Expected Result	• A new subgroup entry appears in the annotations hierarchy. • Setting it as active means any new annotation will be placed in this group.
Result	☐ Pass ☐ Fail
Notes	

TC-10.2: Grouping – Move Leafs

Prerequisites	At least two annotations exist, and a group has been created (TC-10.1 passed).
Steps	1. In the **Annotations tab** list, click the square icon in front of one or more annotations. The square turns green. 2. Set the target group as active (right-click → **“Set Active”**). 3. In the leaf actions panel, click **“Selection: Move”**.
Expected Result	• The selected annotations (green squares) move to the active group in the hierarchy. • They no longer appear under the root group, but appear under the target group.
Result	☐ Pass ☐ Fail
Notes	

11 View Planner

TC-11.1: View Planner – Place Rover

Prerequisites	A surface is loaded. A `rover.xml` file is present in the `Release\InstrumentStuff` folder.
Steps	1. In the actions drop-down, select “**PlaceRover**”. 2. From the rover model selector, choose a rover model. 3. Hold `Ctrl` and click `LMB` on a first point on the surface (rover position – shown in green). 4. Hold `Ctrl` and click `LMB` on a second point on the surface (viewing direction – shown in yellow).
Expected Result	• A rover model appears on the surface at the first picked point. • The rover faces the direction indicated by the second point. • The View Plan entry appears in the **ViewPlanner tab** list.
Result	☐ Pass ☐ Fail
Notes	

TC-11.2: View Planner – Instrument View

Prerequisites	A rover has been placed (TC-11.1 passed).
Steps	1. In the **ViewPlanner tab** list, click the square icon next to a View Plan to select it. 2. From the *Instrument* drop-down, select a camera instrument. 3. Adjust *Sensor (px)* and *Focal (mm)* parameters. 4. Adjust *Pan* and *Tilt* angles. 5. Enable *show footprint*. 6. Click *export footprint*.
Expected Result	• The camera footprint (light gray projection) appears on the surface in the 3D view. • An Instrument View panel appears showing the view from the selected instrument. • Adjusting pan/tilt updates the footprint and instrument view. • Exporting footprint saves a screenshot from the main view, an instrument view screenshot, and a `.svx` metadata file.
Result	☐ Pass ☐ Fail
Notes	

12 GIS View

TC-12.1: GIS View – Load SPICE Kernel

Prerequisites	GIS View is available (application started with GIS mode enabled or via Change Mode). A SPICE kernel file (`.tm`) is available on disk.
Steps	1. Navigate to the **GIS tab** tab in the right panel. 2. Locate the kernel loading section and click the button to browse for or enter the kernel path. 3. Select or enter the path to the `.tm` kernel file. 4. Confirm the selection.
Expected Result	• The kernel path appears in the GIS View settings pane. • A green check mark appears below the path, indicating successful loading. • If loading fails, a red exclamation mark is shown (check PRo3D console for details).
Result	☐ Pass ☐ Fail
Notes	

TC-12.2: GIS View – Observation Settings

Prerequisites	A SPICE kernel is loaded (TC-12.1 passed).
Steps	1. In the *Current Observation Settings* section, set an *Observed Body* (e.g., `MARS`). 2. Set a *Camera Source Body* (e.g., `PHOBOS`). 3. Set a valid *Time* (within the kernel's coverage). 4. Set a *Reference Frame*. 5. Click the **"Reset"** button.
Expected Result	• The camera moves to the location of the Camera Source Body and points toward the Observed Body. • The 3D view updates to show the configured observation scenario. • Entities with proxy geometry enabled are visible in the scene.
Result	☐ Pass ☐ Fail
Notes	

13 Contour Lines and Multitexturing

TC-13.1: Contour Lines

Prerequisites	A multilayer OPC surface (with an `.opcx` file and an elevation layer) is loaded and selected.
Steps	1. With the surface selected, in the surface Properties panel, navigate to the *Textures* selector and choose the elevation layer as the secondary texture. 2. Find the *Contour* section in the surface properties. 3. Enable contour lines by checking the enable checkbox. 4. Set a *Distance* value (line interval). 5. Set *Line Width* and *Line Border* values.
Expected Result	• Contour lines appear on the surface in the 3D view, spaced by the specified distance. • Adjusting the interval changes the spacing between contour lines visibly.
Result	☐ Pass ☐ Fail
Notes	

TC-13.2: Multitexturing

Prerequisites	A surface with an `.opcx` file providing multiple texture and/or scalar layers is loaded and selected.
Steps	1. With the surface selected, find the *Scalars* and *Textures* combo boxes in the Properties panel. 2. Select a different layer from the *Scalars* drop-down (e.g., an elevation or accuracy layer). 3. Locate the *Multitexturing* controls in the surface properties. 4. Adjust the opacity or blending slider.
Expected Result	• The selected scalar/attribute layer is applied as a false-color map on the surface. • A color legend appears on the left side of the 3D view. • Adjusting the blending slider changes how much the additional layer overlays the base texture.
Result	☐ Pass ☐ Fail
Notes	

TC-13.3: Cross Sections

Prerequisites	A surface is loaded and visible. A curtain image (PNG) prepared from an exported profile is available on disk (see the *Cross Sections* section of the Short User Manual).
Steps	1. In **"DrawAnnotation"** mode, choose **"Polyline"** with **"Sky"** projection and draw a polyline along the area to be cut. Use an adequate sampling distance to avoid excessive segment counts. 2. Select the polyline annotation and open its Properties panel. 3. Use **"Export Profile"** to export the profile as a CSV file. 4. Move the camera far away from the cross section so that everything between the annotation and the camera will be clipped. 5. In the annotation properties, trigger **"Create Cross Section"**. 6. Open the *Curtain* settings, set the curtain height and minimum altitude to match the values used when generating the curtain image, and select the prepared curtain image.
Expected Result	• The exported CSV contains `distance,elevation` columns with one row per sampled point. • After creating the cross section, the surface data in front of the cutting plane is clipped away, leaving the cut visible. • The curtain image is mapped onto the cutting plane and aligned with the profile. • The cross section and curtain can be inspected in the 3D view without artifacts.
Result	☐ Pass ☐ Fail
Notes	

14 Surface Comparison

TC-14.1: Surface Comparison – Activate Mode

Prerequisites	Two surfaces with different names are loaded.
Steps	1. Click the main menu icon. 2. Navigate to *Change Mode* → *Surface Comparison*. 3. In the Surface Comparison tab, select one surface from the *Surface1* drop-down. 4. Select the second surface from the *Surface2* drop-down. 5. Press `t` on the keyboard.
Expected Result	• The Surface Comparison interface appears with Surface1 and Surface2 selectors. • Both surfaces can be selected independently. • Pressing `t` toggles visibility between the two surfaces (only one is visible at a time).
Result	☐ Pass ☐ Fail
Notes	

TC-14.2: Surface Comparison – Select Area

Prerequisites	Surface Comparison mode is active and two surfaces are selected (TC-14.1 passed).
Steps	1. In the menu bar above the 3D view, choose "**Select Area**" from the drop-down. 2. Hold `Ctrl` and click `LMB` on the surface to select a location. 3. Observe the sphere indicator. 4. Press `+` to increase the sphere size; press `-` to decrease it. 5. Click the calculator icon button to update measurements.
Expected Result	• A translucent sphere appears at the clicked location. • The `+`/`-` keys change the sphere radius. • After clicking the calculator icon, colored spheres appear within the area, color-coded by distance between the two surfaces (red = large difference, green = small difference). • A color legend is shown on the left side. • Statistics appear in the panel on the right for the selected area.
Result	☐ Pass ☐ Fail
Notes	

TC-14.3: Surface Comparison – Surface Measurements

Prerequisites	Two surfaces are selected in the Comparison tab (TC-14.1 passed).
Steps	1. In the Surface Comparison panel, navigate to the *Surface Measurements* section. 2. From the *Origin* drop-down, select a center calculation method. 3. Click the calculator icon button to update measurements.
Expected Result	• Size measurements along the coordinate axes are listed for each surface. • Differences between the two surfaces are shown below the individual measurements.
Result	☐ Pass ☐ Fail
Notes	

TC-14.4: Surface Comparison – Length Measurements via Bookmarks

Prerequisites	Two surfaces are selected in Surface Comparison mode. A bookmark is available.
Steps	1. Navigate to the **Bookmarks tab** page and select a bookmark (it should be highlighted in green). 2. Switch to **“DrawAnnotation”** mode (`F2`). 3. In the projection selector, choose **“Bookmark”** projection mode. 4. Draw one annotation on Surface1. 5. Press `t` to switch to Surface2. 6. Draw another annotation on Surface2 (same bookmark selected). 7. Return to the Surface Comparison window and click the calculator icon.
Expected Result	• The Bookmark projection mode is available in the DrawAnnotation panel. • Both annotations use the same bookmark as the projection origin. • The *Annotation Length Comparison* section shows the length measurements for both annotations and their difference.
Result	☐ Pass ☐ Fail
Notes	

15 Minerva

TC-15.1: Minerva – Load Data and Product Listing

Prerequisites	A `dump.csv` file is present in the `Release\netcoreapp2.0\MinervaData` folder. PRo3D is started.
Steps	1. Start PRo3D. 2. Navigate to the **Minerva tab** tab in the right panel. 3. Observe the product listing.
Expected Result	• The Minerva tab is visible. • Products are listed, grouped by instrument. • Up to 20 products per instrument are shown in the list. • Products appear as dots/markers in the 3D view after a surface is loaded.
Result	☐ Pass ☐ Fail
Notes	

TC-15.2: Minerva – Query App Filter

Prerequisites	Minerva data is loaded (TC-15.1 passed).
Steps	1. Click the **“Query App”** button (B in the Minerva tab) to open the Query App. 2. Set a *min sol* and *max sol* range. 3. Uncheck some instruments. 4. Click **“ApplyFilter”**. 5. Observe the product listing and 3D view. 6. Click **“ClearFilter”**.
Expected Result	• After applying the filter, only products within the sol range and checked instruments are shown. • The 3D view updates to display only filtered products. • Clicking ClearFilter restores all products.
Result	☐ Pass ☐ Fail
Notes	

TC-15.3: Minerva – Product Selection

Prerequisites	Minerva data is loaded. Surface is loaded.
Steps	1. In the product listing, click on a product name. 2. Observe the Properties panel and the color change in the list. 3. Click the “**FlyTo**” button below the product name. 4. In the actions drop-down, set “**PickMinervaProduct**”. 5. Hold `Ctrl` and click `LMB` on a product dot in the 3D view. 6. Hold `Shift`+`LMB` and drag a rectangle over multiple products in the 3D view.
Expected Result	• Clicking a product name highlights it in green in the list and shows its properties. • FlyTo animates the camera to the product location. • PickMinervaProduct allows selecting a product by clicking it in the 3D view. • Shift-dragging selects multiple products within the drawn rectangle; all are highlighted.
Result	☐ Pass ☐ Fail
Notes	

TC-15.4: Minerva – Pick Linking

Prerequisites	Minerva data is loaded. A surface is loaded.
Steps	1. In the actions drop-down, select “**PickLinking**”. 2. Hold `Ctrl` and click `LMB` to pick a point on the surface. 3. Observe the 3D view.
Expected Result	• Camera frustums are displayed for all products whose field of view captures the selected point. • The frustums are drawn as wire cones/pyramids from each product’s camera location toward the surface.
Result	☐ Pass ☐ Fail
Notes	

16 Command Line Interface

TC-16.1: CLI – Snapshot Rendering

Prerequisites	PRo3D is installed. An OPC surface folder is available. A valid snapshot JSON file (`snapshots.json`) is prepared according to the manual's format.
Steps	1. Open a terminal (Command Prompt or PowerShell) and navigate to the directory containing `PRo3D.Snapshots.exe`. 2. Run: `PRo3D.Snapshots.exe -help` and read the output. 3. Run the basic snapshot command: `PRo3D.Snapshots.exe -opcs MyOPCs\surface\-asnap snapshots.json -out MyImages\` 4. Wait for the rendering to complete. 5. Check the output folder.
Expected Result	• The `-help` flag prints a list of all available arguments. • The rendering process starts and the viewer renders each snapshot. • Rendered image files appear in the specified output folder (`MyImages\`). • The number of output images matches the number of snapshot entries in the JSON file.
Result	☐ Pass ☐ Fail
Notes	

17 RIMFAX

TC-17.1: RIMFAX – Load Traverse

Prerequisites	RIMFAX data (including a `RIMFAX_traverse.json` file and the expected folder structure) is available. A surface is loaded.
Steps	1. Click the main menu icon. 2. Navigate to the *Extra* panel. 3. Click the button to load a RIMFAX traverse file and select the `RIMFAX_traverse.json` file.
Expected Result	• The RIMFAX traverse is loaded and displayed in the 3D view. • The traverse path appears aligned with the rover path on the surface.
Result	☐ Pass ☐ Fail
Notes	

TC-17.2: RIMFAX – Load Surfaces

Prerequisites	A RIMFAX traverse is loaded (TC-17.1 passed). The full RIMFAX folder structure (with OBJ/PNG files) is available.
Steps	1. In the RIMFAX attribute window, click the **"Import Surfaces"** button. 2. Navigate to the root RIMFAX data folder and select it. 3. Observe the 3D view and the Sol panel.
Expected Result	• RIMFAX surface images appear in the 3D view alongside the traverse. • Sol numbers are listed in the Sol panel. • The RIMFAX mode can be changed per sol in the Sol panel.
Result	☐ Pass ☐ Fail
Notes	

18 Keyboard Shortcuts

TC-18.1: Keyboard Shortcuts

Prerequisites	A surface is loaded. At least two surfaces are available for the surface comparison shortcut.
Steps	1. Press `f` – toggle linear texture magnification filtering. Observe the surface texture. 2. Press `p` – show/hide exploration point. Observe the pink explore center dot. 3. Press `t` – toggle visibility between two surfaces selected in the Comparison window. 4. Press `Page Up` – raise navigation sensitivity. Observe faster movement. 5. Press `Page Down` – lower navigation sensitivity. Observe slower movement. 6. Press `Ctrl`+`s` – save the existing scene. Verify the save completes. 7. Press `Ctrl`+`c` – print current camera parameters in snapshot format to the console. 8. Press `Space` – save a waypoint. Observe any console/UI feedback. 9. Press `F1` – switch to Pick Explore Center interaction mode. 10. Press `F2` – switch to Draw Annotation interaction mode. 11. Press `F3` – switch to Pick Annotation interaction mode. 12. Press `F4` – switch to Place Coordinate System interaction mode. 13. Press `Ctrl`+`Z` – undo the last drawing action. Observe the annotation list. 14. Press `Ctrl`+`Y` – redo the previously undone action. Observe the annotation list.
Expected Result	• `f`: Surface texture filtering toggles visually (texture appears smoother/crisper). • `p`: The pink explore center dot appears or disappears. • `t`: Surfaces toggle between visible and invisible (in Comparison mode). • `Page Up`/`Page Down`: Navigation speed changes noticeably. • `Ctrl`+`s`: The current scene is saved without opening a dialog. • `Ctrl`+`c`: Camera parameters are printed in the console output. • `Space`: A waypoint is recorded (feedback in console or UI). • `F1`–`F4`: The selected interaction mode changes in the actions dropdown. • `Ctrl`+`Z`: The last added annotation disappears from the list and the 3D view. • `Ctrl`+`Y`: The undone annotation reappears in the list and the 3D view.
Result	☐ Pass ☐ Fail
Notes	

19 Undo/Redo and Group Default Color

TC-19.1: Undo/Redo for Drawing

Prerequisites	A surface is loaded. At least one annotation has been drawn (e.g., a polyline from TC-3.3).
Steps	1. Draw a new polyline annotation on the surface (**"DrawAnnotation"** mode, **"Polyline"**, finish with `Enter`). Note its name in the **Annotations tab** list. 2. Press `Ctrl`+`Z` (Undo). 3. Observe the **Annotations tab** list and the 3D view. 4. Press `Ctrl`+`Z` again to undo a second annotation (if one exists). 5. Press `Ctrl`+`Y` (Redo). 6. Observe the **Annotations tab** list and the 3D view. 7. Draw another annotation, then in the **Annotations tab** list click **"remove"** to delete it. 8. Press `Ctrl`+`Z` to undo the deletion. 9. Observe the **Annotations tab** list.
Expected Result	• After `Ctrl`+`Z`: The last drawn annotation is removed from both the **Annotations tab** list and the 3D view. The undo is immediate with no dialog. • Multiple `Ctrl`+`Z` presses step back through the annotation history one action at a time. • After `Ctrl`+`Y`: The undone annotation reappears in the list and the 3D view, restoring it exactly as it was. • Undoing a deletion (**"remove"**): The deleted annotation is restored in the list and the 3D view. • When there is nothing left to undo/redo, additional presses of `Ctrl`+`Z`/`Y` have no effect (no error or crash).
Result	☐ Pass ☐ Fail
Notes	

TC-19.2: Group Default Color

Prerequisites	A surface is loaded. At least one annotation group exists in the **Annotations tab** panel (the "root" group is always present).
Steps	1. Navigate to the **Annotations tab** panel. 2. In the group hierarchy, locate the active group (e.g., "root" or a subgroup created in TC-10.1). 3. Find the color picker labeled **Default color for new annotations in this group** shown in the group's action area. 4. Change the default color to a clearly distinct color (e.g., bright red). 5. Make sure this group is set as active (right-click → **"Set Active"**). 6. Switch to **"DrawAnnotation"** mode (`F2`) and draw a new polyline annotation on the surface. 7. Observe the color of the newly drawn annotation in the 3D view and its color in the Properties panel. 8. Change the group's default color to a different color (e.g., blue). 9. Draw another annotation. 10. Observe that each new annotation uses the group's current default color at the time of drawing.
Expected Result	• A color picker is visible in the group's context area in the **Annotations tab** panel. • The first new annotation is drawn in the red color that was set as the group default. • The annotation's *Color* field in the Properties panel shows the same red color. • The second new annotation is drawn in blue. • Existing annotations are **not** recolored when the group default color changes; only newly created annotations inherit the current default color. • The default color is persisted when the scene is saved and reloaded (TC-1.4/TC-1.5).
Result	☐ Pass ☐ Fail
Notes	

Defect Log

Use this table to record any defects found during testing.

ID	Test ID	Summary	Steps to Reproduce / Observation	Severity
D-01				
D-02				
D-03				
D-04				
D-05				
D-06				
D-07				
D-08				

Severity levels: Critical (blocks testing), High (major feature broken), Medium (feature partially broken), Low (minor issue / cosmetic).